A ONE STEP DEVISSAGE IS PRESERVED BY AN ARBITRARY AFFINE BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros; no macro was invented or redefined.

Editorial note. The same-chapter referenced definition is located at its \label line inside the pinned chapter file; the nine cross-chapter permanence statements are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- One step devissage of a finite type module over a point. Same chapter, Definition environment labelled definition-one-step-devissage, source line 780 (Stacks tag 05H4). Supplies the list of properties that a one step devissage consists of and that the counted proof has to verify one by one after base change.
- Base change behaviour of closed immersions, finite presentation, finiteness, smoothness, geometric irreducibility of fibres and relative dimension. Morphisms, Lemmas morphisms-lemma-base-change-closed-immersion (01QR), -finite-presentation (01TS), -finite (01WL), -smooth (01VB), -relative-dimension-d (02NK) and More on Morphisms, Lemma more-morphisms-lemma-base-change-fibres-geometrically-irreducible (0555); invoked at proof lines 903-909. Supply the list of permanence statements; assembling them into the base-changed tuple is the counted step.
- Base change of pushforward along a finite morphism and dimension of supports on fibres. Cohomology of Schemes, Lemma coherent-lemma-affine-base-change (02KG, proof line 912) and Morphisms, Lemma morphisms-lemma-dimension-fibre-after-base-change (02FY, proof line 916). Supply that i'_*G' is F' and that the two fibre supports have the same dimension; both are quoted inputs for the two non-formal checks of this proof.

Source reference. Original definition 05H4: sources/stacks-flat/pinned-original-source.tex, source lines 780–780. Same-chapter pointer to the definition named in the statement at line 881; it lists the properties whose base change behaviour the counted proof verifies, and that verification is retained verbatim in the delivered body.

Source reference. Original lemma 01JQ: sources/stacks-chow/tags.txt, source lines 2023–2023. Cross-chapter standard prerequisite invoked once at proof line 895 for affineness of the base-changed schemes.

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 01QR: sources/stacks-chow/tags.txt, source lines 2276–2276. Cross-chapter permanence statement cited at proof lines 903-904 for the closed immersion i'.

Source reference. Original lemma 01TS: sources/stacks-chow/tags.txt, source lines 2385–2385. Cross-chapter permanence statement cited at proof line 904 for finite presentation of i'.

Source reference. Original lemma 01WL: sources/stacks-chow/tags.txt, source lines 2486–2486. Cross-chapter permanence statement cited at proof line 905 for finiteness of pi'.

Source reference. Original lemma 01VB: sources/stacks-chow/tags.txt, source lines 2440–2440. Cross-chapter permanence statement cited at proof line 906 for smoothness of the base-changed morphisms.

Source reference. Original lemma 02NK: sources/stacks-chow/tags.txt, source lines 3483–3483. Cross-chapter permanence statement cited at proof line 907 for preservation of relative dimension n.

Source reference. Original lemma 0555: sources/stacks-chow/tags.txt, source lines 6663–6663. Cross-chapter permanence statement cited at proof lines 908-909 for geometric irreducibility of the base-changed fibres.

Source reference. Original lemma 02KG: sources/stacks-chow/tags.txt, source lines 3371–3371. Cross-chapter standard prerequisite invoked once at proof line 912 for the identification i'_*G' = F' via base change of pushforward along the finite i.

Source reference. Original lemma 02FY: sources/stacks-chow/tags.txt, source lines 3209–3209. Cross-chapter standard prerequisite invoked once at proof line 916 for equality of the dimensions of the two fibre supports; the displayed identification of those supports is retained in the counted span.

Lemma 0.1. *Let S, X, $\mathcal{F}$, s be as in Definition 05H4. Let $(Z, Y, i, \pi, \mathcal{G})$ be a one step dévissage of $\mathcal{F}/X/S$ over s. Let $(S', s') \to (S, s)$ be any morphism of pointed schemes. Given this data let X', Z', Y', i', π' be the base changes of X, Z, Y, i, π via $S' \to S$. Let $\mathcal{F}'$ be the pullback of $\mathcal{F}$ to X' and let $\mathcal{G}'$ be the pullback of $\mathcal{G}$ to Z'. If S' is affine, then $(Z', Y', i', \pi', \mathcal{G}')$ is a one step dévissage of $\mathcal{F}'/X'/S'$ over s'.*

Proof. Fibre products of affines are affine, see Schemes, Lemma 01JQ. Base change preserves closed immersions, morphisms of finite presentation, finite morphisms, smooth morphisms, morphisms with geometrically irreducible fibres, and morphisms of relative dimension n, see Morphisms, Lemmas 01QR, 01TS, 01WL, 01VB, 02NK, and More on Morphisms, Lemma 0555. We have $i'_*\mathcal{G}' \cong \mathcal{F}'$ because pushforward along the finite morphism i commutes with base change, see Cohomology of Schemes, Lemma 02KG. We have $\dim(\mathrm{Supp}(\mathcal{F}_s)) = \dim(\mathrm{Supp}(\mathcal{F}'_{s'}))$ by Morphisms, Lemma 02FY because

$$\mathrm{Supp}(\mathcal{F}_s) \times_s s' = \mathrm{Supp}(\mathcal{F}'_{s'}).$$

This proves the lemma. □